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Machine Learning Approaches for IoT-Data Classification

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Abstract: The emergence of concepts like Data Science, Big Data, Machine Learning, IoT in recent years have added the potential of research in today's world which is flooded with data. The use of IoT devices, sensors, etc. which are collecting data continuously is putting a lot of pressure on the existing IoT network which is resource-constrained. And hence pointing towards the potential of research for such kind of resource-constrained environment. In this paper, the focus is on the classification of data at the device level, edge/fog level and cloud level using machine learning techniques. The data analysis for the same will be done by using Python language. As the data which is coming from different devices is vast and is of variety, therefore it becomes very important to choose the right technique for the particular type of data for classification. This will help in optimizing the data at the device, edge/fog level for better performance of the network in the future.

Keywords: Data analytics, Python, IoT, Classification

1. Introduction

In today's world data is getting generated by the human population using smart devices every single minute. Researchers estimate that over 3 million new devices are connected to the internet each month. Researchers also estimate that in the next four years, there are going to be over 30 billion connected devices worldwide. [1] By the year 2020, nearly twenty to thirty billion electronic devices will be connected as part of the Internet of Things (IoT) [2]. Internet of Things (IoT) is becoming a highly anticipated fuel for big data revolution by interconnecting smart devices in cities i.e. connecting all the vehicles, appliances, industrial networks, health-care services, etc. to the internet [3]. To improve the quality of living style, the internet of things (IoT) is undoubtfully the solution and key technology for the current world having high capabilities for the future. But at the same time, it also puts a lot of pressure on the existing method for communication at the Internet of Things (IoT) network. The Internet of Things (IoT) is going to exceed around 4.40 zettabytes amounting to 10 percent of the global "digital universe" by 2020, which was only 2 percent in 2012-2013.

For communication technologies, a solution which tends to improve by itself for the future is highly required. Because all the existing infrastructures are not going to work for the future after looking at the growth rate of the Internet of Things (IoT) data.

All types of data which are generated by the IoT devices are assumed to be uploaded and stored to the cloud viz. IOT/cloud integration. To extract useful knowledge cloud-based data analysis is done and results are generated for IoT data. This approach is not sufficient for the majority of applications having a large amount of redundancy in the collected data. As the Internet of Things (IoT) network is alarming because of the tremendous increase in the growth rate of the IoT data, the existing Machine Learning algorithms are to be redefined in the resource-constrained network for an efficient and everlasting solution.

Classification and Regression are the techniques of Machine Learning approaches, which fall under the category of Supervised Learning. The machine learning algorithms have the capability of classifying the data in a particular domain which helps predict the future response or helps in the decision support system. As in the case of the Internet of Things (IoT) network which is flooded with the IoT data and is lacking the resources at different Levels i.e. device level, Edge/fog Level. Therefore, the classification technique is the best possible way of classifying the data at each level, so that the data is filtered at the very first level i.e. device level. If the data is classified at the device level then the resource constraints like Memory, Processing Unit, Communication Unit, and Power Source will not be a barrier for the volume, velocity, and variety of the IoT data in the future.

2. Literature Survey

The amount of data in an IoT network is so high that the existing framework and presently available algorithms are not sufficient for processing the IoT data. Therefore, the improvement in the present IoT network is a big Challenge and has high importance for the future of IoT data Analytics. Mohammad Saeid, Mohammadreza Rezvan, Mohammadamin Barekatain, Peyman Adibi, Payam Barnaghi d and Amit P. Sheth have surveyed about the use-case of different machine learning algorithms that have been recently used for processing of IoT data in an IoT network [20].

The data shows that the use cases like smart traffic, smart health, smart environment, etc. are processed at the edge, edge/cloud and cloud respectively. The smart air controlling, smart public places and smart human activity has historical data and is processed at cloud/edge levels.

The type of data for the smart cities and the location of its processing viz Edge or Cloud is different in the IoT Network. The most of the processing is done at the edge and the cloud level and most of the researchers have shown the same type of Data characteristics [21-32], which further leads us towards the IoT data Analytics at all the levels i.e., cloud, edge/fog, and device level (device-head cluster). The data captured by the device is forwarded to the above level, as the resources at the device level are limited therefore researchers are restricted for data processing at the device level. The Typical IoT Node is shown below.

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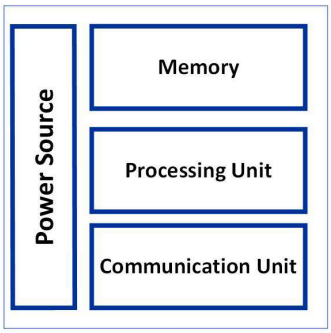

Fig. 1- Resources of a Typical IoT Node [4]

In a smart environment, the sensing and the collection of data are performed by the physical nodes that deals with the communication nodes of the IoT. In a way, like gateways are responsible for connecting to other smart things, the network communication nodes are responsible for Data aggregation, conversion, and transmission.

2.1 IoT Architectures

The IoT architecture comprises of the following layers:

- The Perception Layer.
- Network Layer.
- Middle Layer.
- Application Layer and
- The Business Layer.

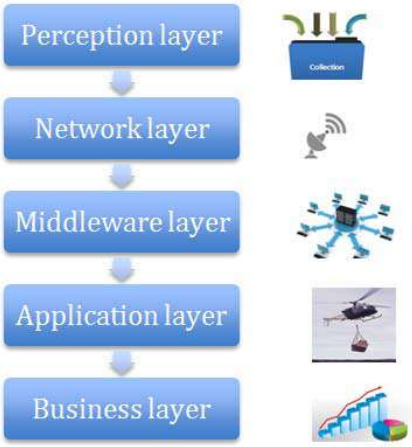

Fig. 2- Basic Architecture of IoT

Perception Layer: This layer has the capability of data collection and is responsible for collecting and identifying the data. After that, the data is analyzed and is accessed by an IoT application to give service to the users. In an IoT network, each device is divided into many fields of mini-profile that can be similar or different from the other IoT devices present in the IoT environment. This information is going to be of great value to control the shared data within its cluster groups or outside the cluster.

***Networking Layer*:** Network Layer connects all entities and permits the sharing of data between different connected entities of IoT. It is capable of aggregating data of existing IT infrastructures like business systems, transportation systems, power grids, tending systems, ICT systems, etc. Services provided by Layers of Architecture usually develop a heterogeneous network that is more complex.

When we talk about the IoT networking layer, researchers have to take certain considerations like energy efficiency in the network, resource availability, QoS requirement, Data and Signal Processing and most importantly security and privacy [6].

Middleware Layer: has two important tasks i.e. management of Service which is also called service management and the second task is storing the information present in the lower layers of the network. This layer has more capabilities like decision making based on computational results, retrieving information, processing and computing the information.

Application Layer: The information that is present in the Middleware Layer is utilized by the application layer to manage new applications.

The examples of the IoT new applications are smart health, smart glasses, smart independent living, smart transportation, smart home, etc.

Business Layer: IoT applications and management services fall under this very layer. It can make or describe Practical reality or business trends of the data with the help of flowcharts, graphs, models, etc.

The processed data which it receives from the lower levels of architecture is processed effectively for Data Analytics, which makes it easier for the functional managers or executives to do further accurate analysis. The analysis is further required for important decisions or plans.

2.2 Functions of IoT Architecture

There are three essential functions of an IoT architecture which are given below

- Aggregation of data and sensing.
- Utilizing data and
- Communicating data

In a smart environment, the sensing and the collection of data are performed by the physical nodes that deals with the communication nodes of the IoT. In a way, like gateways are responsible for connecting to other smart things, the network communication nodes are responsible for Data aggregation, conversion, and transmission. The sensed data is then accessed by IoT applications to perform for various services to the users.

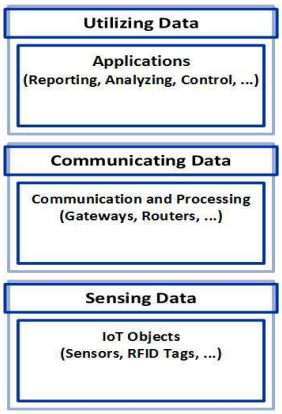

Fig. 3. Basic Functions of IoT [4]

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A large amount of data is been generated by the ever-increasing number of devices that collect processes and uploads data to the cloud. [1] Internet of things is considered as a key technology to improve the quality of life of economic growth and employment at the global level. One of the main challenges for IoT to cloud integration is contemporary communication technologies. The IoT cloud integration includes uploading and storing data generated by IoT devices therefore IoT data have to be processed by cloud-based data analysis to extract useful information.

In the majority of the cases, we found that the tendency of the collected data is very high therefore it becomes necessary to pre-process the data for analysis and storage purposes. Due to the growth of IoT, it becomes necessary to redesign IoT communications [5]

As shown in fig. 3 the basic functions of an IoT system. The basic architecture involves three functions that are an aggregation of data, sensing, utilizing data and communicating the same for various services at the application level. The data collection and sensing are performed by physical IoT. The network communication nodes are responsible for data aggregation and transmission to other nodes or networks.

The censored data is then accessed by IoT applications to provide various services to the users. It includes a communication unit, processing unit, memory and a power source However IoT networks comprises of many complex heterogeneous components.

The Typical IoT nodes have many resource constraints, therefore to envision real-world IoT systems is difficult without the inclusion of cloud platforms or some powerful devices including smart gateways, fog devices, etc. The resource limitation should be taken care of at both the levels hardware as well as software level.

In addition to the source limitations, there are many constraints due to the characteristics of IoT nodes which include the behaviour of the network, constraint at the application level, etc. These constraints are mostly applications dependent and commonly found in pervasive applications of IoT.

Some of the applications of IoT are

- Remote Monitoring of Soil Parameters
- Environment monitoring
- Monitoring of green energy system.
- Water Monitoring.
- Disaster Monitoring.
- Remote Monitoring Patients.

Etc.

3. Machine Learning Approaches in Internet of Things (IoT)

Machine learning algorithm plays a vital role in the analytics of IoT data. [33][34]. As a subset of Artificial Intelligence, it has a great significance in the IoT network. With the increasing amount of data, day by day produced or generated by our own devices is of no use if not processes properly and utilized for data analytics like Big Data Hadoop, MATLAB, Bluemix, etc. Similarly, a huge amount of data is generated in the network of IoT. This network is similar to the wireless network system in terms of architecture to accept a few things. The layers in the architecture require the Machine learning algorithms for different levels separately. The common machine learning algorithms that are presently being utilized by different researchers are Naivie bayes, k- Nearest Neighbor, K- means, PCA, SVM, Random Forest, etc. [7] These algorithms have different functions and capabilities of accessing and processing the data.

The k-means algorithms have significant existence among the group of algorithms present today. But it has one big problem i.e. the researcher should be confident about the value of the parameter k, otherwise, the algorithms will lead towards null or towards a bad decision or output. Mohammad Saeid et al. have presented the algorithm for K-means where the cluster center S_k is learning to be optimal for the data and the assignment of the data πnk.

The table 1 below is a comparison of different machine learning algorithms based on their advantages and disadvantages.

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Table 1- Comparison of different machine learning algorithms based on their advantages and disadvantages

Algorithm	Use Case for IoT	Advantages	Disadvantages
K-means [7],[8]	Traffic and Air Control monitoring, making home as smart home, so will be cities as Smart Cities.	Better for huge variables and also produces tighter clusters	K- value difficult and difficult for data of different densities
Naive Bayes [10],[11]	Agriculture data analysis, All citizens as smart citizens.	Simple to implement. Needs less training data	Assumption can be wrong.
K-Nearest Neighbour [12]	Smart Citizen	It will be helpful even if the training data is noisy.	The value for k must be known as accurately.
Support Vector Machine (SVM) [13],[14]	Classify data, real-time Prediction.	Data can be visualized in more than 2 dimensions. Decision taking system is nearly perfect.	Selection of a good kernel value is not easy, And also, if the data set is large then it will take more time for training.

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PCA [15-19]	Monitoring Public Places, Useful in Fault Detection.	Lack of redundancy, Reduced complexity in images.	The covariance matrix is difficult to evaluate and even the simplest invariance could not be captured.
Neural Network [8]	Low energy consumption, Healthcare Analysis, Forecasting, Useful with redundant data also.	Ability to learn and model non-linear and complex relationships. ANNs can generalize.	Parameters are hard to interpret. It becomes very complex sometimes. Needs a large dataset.
Random Forrest [8]	l-Banking, Stock Market, Medicine, and E-Commerce	considered as good for takings thousands of input variables.	Noise in the datasets

3.1 Machine Learning Advancements

In earlier days if someone would have thought about machines or algorithms that learn by itself then nobody would have believed. Today we have machine learning algorithms that learn by itself with the help of some parameters or training data. The most important branch of Artificial intelligence i.e. machine learning has advanced tremendously in recent years. Machine learning has four sub-sets i.e. supervised learning, unsupervised learning, semi-supervised learning, and Reinforcement learning.

The supervised learning means that the algorithm has some labelled data which is used by the algorithm/machine to predict the future. Therefore, also known as a predictive model.

Unsupervised learning has no labelled data but its learning process fully depends on the selection of the parameters on which the comparison or the decision-making system works. Therefore, also known as a descriptive model.

The reinforcement learning will use none of the above methods except the learning after the failure method. It detects the failure pattern and prevents the system from using it again.

The table below shows the recent advancements in machine learning algorithms by different researchers.

Table 2. Advancements in the machine learning algorithms by different researchers.

Author(s) name (year)	Algorithms /Techniques	Summary
J. L. Berral-G arcia (2016) [40]	SVM, ANN, K-means, DBSCAN, Decision tree Algorithms.	Classification, prediction and modeling survey was observed and completed using machine learning
Qui, Wu. Q, Ding. G, Feng 2016 [41]	Support vector machine, Regression, Neural Networks, Principal Component Analysis.	Using big data processing, A survey of regular and uncommon algorithms was done.
Bokhari, Zeyauddin, and Siddiqui (2016) [42]	ANN, SVM, PCA, Naive Bayes.	They proposed a new model for storage and Analysis for Big IoT data.
Wu, W. Cheng, Hoffman and Wang (2016) [43]	Principle Component Analysis, Regression	In this paper, health care analytics was introduced.
R. Bendre, C. Thool and V. Thool [44	MapReduce, Linear regression	They developed a model using Regression and mapping and Reducing Technologies.

4. Conclusion

The Data analysis at different levels of IoT-Fog integration has been necessitated by the fact that most of the IoT devices are resource Constrained. Data management has to be the top priority for such a system integration to optimize resource utilization. Each layer/device level has to be identified in an autonomous environment for data analysis as per the capability by using machine learning at each layer/device level. IoT data Analysis in an IoT network for optimizing resource utilization at each level of the IoT network. This can be achieved by minimizing the data flow in the network. Improving data management at each level designing efficient machine learning algorithms for each level. The implementation of the data analysis algorithms and machine learning algorithms will be done by using Python and MATLAB. The data for the analysis will be acquired in Realtime or data sets from the deployed IoT networks for different applications.

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